

Stat 486 Course Project

Due: Thursday May 7, 2026

1 Introduction

For this project, we will investigate data coming from the National Longitudinal Study of the Youth 1997 (NLSY97). Please take a look <https://nlsinfo.org/content/getting-started/introduction-to-the-nls>. I have downloaded some data using the NSL Investigator tool <https://www.nlsinfo.org/investigator/pages/home>; I encourage you to take a look.

You can find the data in the file STAT486_NLSY97.csv. The variables are (some variables are taken directly from the source data while others are derived from others based on certain criteria):

- ID = unique identifier
- Demographics: sex, date of birth, age in 1997, race
- Family: biological father education, biological mother education, parents' income, number of siblings, responded lives with both biological parents (intact = 1), urban / rural at age 17, geographic region at age 17
- School: 8th grade grades, gpa in high school (coarse summary), whether respondent agrees that teachers are good, whether respondent agrees that school is safe, most recent highschool is college prep,
- Other: delinquency index, marriage status by age 17, percent of peers who plan to go to college, ASVAB percentile rank (some measure of ability)
- College completion by age 25 (collcomp25)

We are currently working on constructing a relevant outcome variable in this context. The construction of the data has not undergone sufficient scrutiny; it is meant to be used solely in this course for practice. I will update the dataset in case anything needs to be modified.

2 Goal

There are two main goals: 1) to understand the characteristics of students who are less vs more likely to complete college, and 2) whether completing college has a positive effect on an outcome of interest (more on this later). You will split into **3/4-person teams**,

analyze the data and **summarize your findings in a two-page** report to be submitted by Thursday May 7, 2026. A good report will contain 1) informative figures and 2) a sound statistical approach to answer the scientific question. We will have a few months to work on this— we hope it will be a challenging but rewarding project, let's be creative!